Determination of microbial transglutaminase in meat and meat products.

Transglutaminase is an enzyme that can be used to cross-link pieces of meat, fish or meat products. The resulting product gives the optical impression of an intact chunk of meat. The usage of transglutaminase as a food additive is permitted in some countries. However, its utilisation has to be declared to ensure transparency for consumers. This paper describes two orthogonal analytical methods suited for the detection of technological relevant transglutaminase concentrations (around 25 mg pure enzyme in 1 kg of product) in meat and meat products. The mass spectrometry-based approach relies on a previous digestion with Achromobacter lyticus protease and LC-MS/MS separation and detection. Sufficient selectivity was obtained by monitoring four different peptides. The orthogonal (complementary and independent), ELISA-based approach relies on two commercially available bacterial transglutaminase-specific antibodies, combined to a sandwich ELISA. The two methods were tested by analysing some 60 samples obtained from the market.